

Venzini

Requirements Elicitation Document

An AI-Assisted Yield Management Platform for Data-Driven Optimization in the Petroleum Industry

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1. Executive Summary

Project purpose

Venzini is an AI-assisted web platform that helps petroleum distributors forecast fuel demand and plan deliveries across their network of stations. It applies yield management principles: getting the best overall result from limited resources such as tankers, drivers, fuel and time.

An operations manager selects a station and a fuel type, such as Petrol 95 or Diesel, and sees a 7-day demand forecast based on historical sales, previous orders, current tank levels, weather, local events and seasonal patterns. Each forecast shows how confident the platform is and which factors influenced the prediction, and uncertain forecasts are clearly flagged. The platform also shows daily forecast totals for the whole network and for each warehouse.

Using these forecasts, each station's storage capacity and safety-stock level, and the operational resources uploaded by the dispatcher (tankers with their compartments, drivers and warehouse stock), Venzini recommends how many litres to deliver to each station, which stations to prioritize, and an efficient multi-stop route for every available tanker, together with the fuel cost of each route in US dollars. Each station is served from its closest warehouse.

The manager reviews the plan on an interactive map. They can reorder stops or move stations between tankers by drag and drop, adjust delivery quantities within safe limits, and immediately see the impact on time, distance, cost and risk. They approve or reject each action or the whole plan; rejecting a station triggers a new optimal plan. Nothing is executed without the manager's approval, and every decision is recorded.

The platform is organized around four main screens: a network overview with a map and stockout-risk alerts; a station-demand page with forecast charts and contributing factors; a delivery planner with recommended quantities and route controls; and a recommendations page where managers review, adjust, approve or reject each action. The first MVP is a clickable prototype with fake data; version 1 then runs on sample data with simulated forecasts and routes. Venzini does not set pump prices or control fuel systems.

Business problem

Distributors plan deliveries with limited visibility of future demand. This leads to three costly problems: stations running out of fuel, excess inventory tying up money and storage, and unnecessary or inefficient delivery trips.

Target users

- **Operations manager:** plans, edits and approves deliveries, and manages station records.
- **Dispatcher:** uploads the operational resources available each day (tankers, drivers, warehouse stock) and records actual trip times.
- **System admin:** manages accounts, roles and stations.

Expected value

- Fewer stockouts and less excess inventory.
- Fewer kilometres and lower tanker fuel cost.
- Faster, more confident planning, with every recommendation explained and every decision traceable.
- Full flexibility: the manager can change the stop order, tanker assignment and quantities of the plan and immediately see the impact.

2. Stakeholder Identification

Stakeholder	Role	Needs	Influence	Main Concerns
Operations manager	Primary user. Plans deliveries, manages station records and approves or rejects recommendations.	Accurate forecasts, clear priorities, flexible route editing, visible risks and costs.	High	Trusting the forecasts; being blamed for stockouts; losing control to the system.
Dispatcher	Primary user. Uploads the operational resources available for each delivery day and records actual trip times.	A quick way to record tankers, drivers and warehouse stock; certainty the manager plans with current data; forecast totals per warehouse.	Medium	Outdated or wrong data being used for planning.
System admin	Primary user. Manages accounts, roles and stations.	Simple user and station management; correct access control.	Medium	Unauthorized access; misconfigured roles.
Tanker drivers	Secondary. Execute the approved routes. No platform access in any version.	Feasible routes and stop orders.	Low	Unrealistic routes, overloaded tankers, late changes.
Warehouse / depot team	Secondary. Loads tankers with approved quantities.	Correct and final quantities per tanker.	Low	Last-minute changes; quantities above warehouse stock.
Fleet manager	Secondary. Manages tanker availability and consumption.	Tankers used within capacity and efficiently.	Low	Overloading; extra kilometres and fuel cost.
Distributor management / owners	Business. Fund and sponsor the platform.	Fewer stockouts, less excess inventory, fewer trips, lower cost.	High	Return on investment; operational risk from wrong recommendations.
Finance team	Business. Follows delivery and inventory costs.	Visible route costs and savings.	Medium	Hidden costs from manual overrides.
End customers (drivers at the pumps)	External. Indirectly affected.	Stations that always have fuel.	Low	Stockouts at stations.
Regulators	External. Set fuel storage and transport rules.	Operations that respect storage and transport rules.	Medium	Overfilled tanks or overloaded tankers.
Data providers	External. Supply weather and local events data (simulated in v1). A dedicated platform role is planned for a later version.	Defined data needs.	Low	Unclear data formats and usage.
Developer (Mohamed Zouari)	Project team. Designs and builds Venzini.	Clear, stable, testable requirements.	High	Scope creep; ambiguous rules; one-person workload.
Station managers (future)	Future stakeholder. Dedicated interface planned in a later version.	Visibility on their own deliveries.	Low (v1)	Not involved in v1.

3. User and System Context

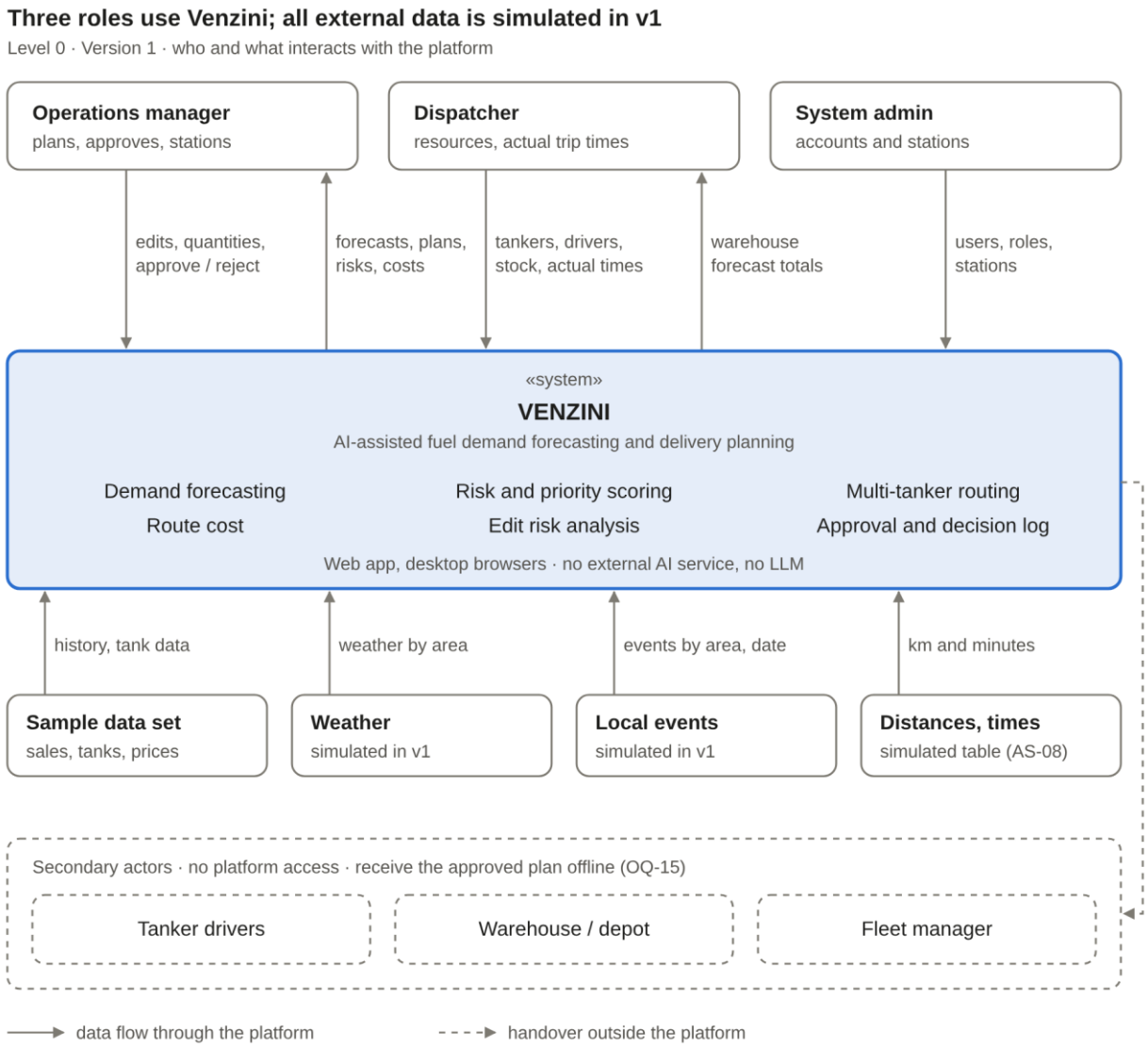


Figure 1 – System context diagram

Primary actors

- Operations manager, dispatcher, system admin.

Secondary actors

- Tanker drivers, warehouse / depot team and fleet manager. They are affected by the plan but have no access to the platform.

External systems

- Weather data source (simulated in v1).
- Local events data source (simulated in v1).
- Map and distance / travel-time data (map service or simulated table, AS-08).
- Sample data set: sales history, stations, tanks (v1).

External organizations and services

- Regulators: rules on fuel storage and transport.
- Data providers: weather and events data in future versions, through a dedicated data-provider role.

Main interactions

- Dispatcher → Venzini: uploads operational resources (tankers with compartments, drivers, warehouse stock) and actual trip times.
- Venzini → Manager: forecasts, priorities, routes, costs, risks and explanations.
- Venzini → Dispatcher: forecast totals per warehouse and fuel type.
- Manager → Venzini: edits routes and quantities, manages stations, approves or rejects recommendations.
- Admin → Venzini: manages users and stations.
- Approved plan → drivers and depot: handed over outside the platform (OQ-15).

4. Business Goals

- **BG-01 – Reduce stockouts at stations.**
  - **Rationale:** Stockouts cause lost sales and unhappy customers.
  - **Success indicator:** Fewer simulated stockout events per period compared with a baseline plan made without Venzini. Target value: to be defined (OQ-11).
- **BG-02 – Reduce excess inventory.**
  - **Rationale:** Overstock ties up money and storage capacity.
  - **Success indicator:** Lower average litres above the target stock level compared with the baseline. Target: OQ-11.
- **BG-03 – Reduce unnecessary delivery trips and route fuel cost.**
  - **Rationale:** Every trip costs fuel, time and tanker use.
  - **Success indicator:** Lower total kilometres and lower route fuel cost per delivered litre compared with the baseline. Target: OQ-11.
- **BG-04 – Keep managers in control with explainable recommendations.**
  - **Rationale:** Managers must trust and own every decision.
  - **Success indicator:** 100% of executed actions have a logged manager approval; 100% of forecasts, priorities and risks show their reasons.
- **BG-05 – Provide flexible planning that adapts to each day's resources.**
  - **Rationale:** Stakeholder statement: the app must be very flexible with users' needs.
  - **Success indicator:** The manager can reorder stops and move stations between tankers and see the impact without leaving the map.

5. User Needs

ID	Stakeholder	Need
UN-01	Operations manager	See at a glance which stations are at risk of running out.
UN-02	Operations manager	Know the expected demand for the next 7 days per station and fuel type.
UN-03	Operations manager	Know how reliable a forecast is and what influenced it.
UN-04	Operations manager	Know which stations to serve first and why.
UN-05	Dispatcher	Record the operational resources available for the delivery day (tankers, drivers, warehouse stock).
UN-06	Operations manager	Get recommended quantities and routes that use all available tankers.
UN-07	Operations manager, Finance team	Know what each route costs in tanker fuel.
UN-08	Operations manager	Read routes on a clean map, with details only when needed.
UN-09	Operations manager	Change the plan: stop order, tanker assignment and delivery quantities.
UN-10	Operations manager	Understand the risks and costs of every change before deciding.
UN-11	Distributor management	Be sure nothing happens without human approval.
UN-12	Distributor management	Trace who decided what, and when.
UN-13	System admin	Control who can access which functions.
UN-14	System admin, Operations manager	Maintain the list of stations.
UN-15	Distributor management	Keep pricing and fuel-system control outside the platform.
UN-16	Developer	Build and demonstrate the platform without real distributor data.
UN-17	Dispatcher, Warehouse / depot team	Know the total expected demand per warehouse and fuel type without adding up station forecasts.
UN-18	Operations manager	Compare estimated and actual trip times to improve planning over time.

6. Functional Requirements

Each requirement is atomic and traces back to a user need (UN), the Requirements Specification (RS), stakeholder input or a documented assumption (AS). Priority: Must = version 1 core; Should = rest of version 1. The first MVP is a clickable prototype with fake data that covers the Must screens.

ID	Requirement	Source	Priority	Acceptance Criteria
Access and roles				
FR-01	The system shall require users to log in before accessing any page.	UN-13; RS NFR7	Must	Opening any page without being logged in redirects to the login page.
FR-02	The system shall support three roles: operations manager, dispatcher and system admin.	UN-13; RS FR19, NFR7	Must	A test account of each role sees only the functions allowed for that role (BR-02, BR-03, BR-22).
FR-03	The system shall allow only operations managers to approve or reject recommendations.	UN-11; RS NFR7	Must	Dispatcher and admin accounts cannot see or use approve/reject controls.
FR-04	The system shall allow only dispatchers to upload or update resources and actual trip times.	UN-05; RS NFR7	Must	Manager and admin accounts cannot save resource data.
User and station administration				
FR-05	The system shall allow the system admin to create, edit and deactivate user accounts and assign their role.	UN-13; RS FR19	Must	A new account can log in with its role; a deactivated account cannot log in.
FR-06	The system shall allow the system admin and the operations manager to create, edit and deactivate station records.	UN-14; RS FR20	Should	A new station appears on the map; a deactivated station no longer appears in plans.
Resource management				
FR-07	The system shall allow the dispatcher to upload the tankers available for a delivery day, with each tanker's compartment sizes and total capacity in litres, and its home warehouse.	UN-05; RS FR15; stakeholder input	Must	Saved tankers, compartments, capacities and home warehouses are shown on the planning page for that day.
FR-08	The system shall allow the dispatcher to upload the drivers available for a delivery day.	UN-05; stakeholder input	Must	Saved drivers are shown on the planning page and are used when tankers are assigned.
FR-09	The system shall allow the dispatcher to upload the fuel stock of each warehouse, per fuel type, in litres.	UN-05; RS FR15; stakeholder input	Must	Saved stock values per warehouse are shown on the planning page.
FR-10	The system shall allow the dispatcher to upload other operational resource data once it is defined.	UN-05; stakeholder input; OQ-20	Should	A newly defined resource type can be uploaded and is visible on the planning page.
FR-11	The system shall allow the dispatcher to update the operational resources of a delivery day until its plan is decided (every action approved or rejected).	UN-05; RS FR16; AS-05	Must	Updates are accepted before the plan is decided and refused after.
FR-12	The system shall use the most recently saved resources when generating a delivery plan.	UN-05; RS FR17	Must	After an update, a newly generated plan uses the new tanker count and stock.
FR-13	The system shall display the date, time and user of the last resource update.	UN-05; RS FR18	Must	The planning page shows the correct user and timestamp after an update.

ID	Requirement	Source	Priority	Acceptance Criteria
FR-78	The system shall calculate the estimated departure time, arrival time at each stop and return time of each trip, using travel times and an unloading time that depends on the delivered quantity.	UN-06; BR-23, BR-25, BR-29	Must	For a test route, the estimated return equals the departure plus all travel and unloading times.
FR-79	The system shall allow the dispatcher to record the actual departure and return time of each trip.	UN-05, UN-12; BR-29	Must	Recorded times set the trip to “departed”, then “returned”; other roles cannot record them.
FR-84	The system shall store and show the difference between the estimated and the actual duration of each trip.	UN-18	Should	A trip estimated at 8 h and recorded at 8 h 45 min shows +45 min (+9.4%).
Network overview				
FR-14	The system shall display all active stations on a map.	UN-01; RS FR1	Must	The number of markers equals the number of active stations in the data set.
FR-15	The system shall display a stockout-risk alert for each station and fuel type with a high risk level.	UN-01; RS FR2	Must	A sample station set to high risk shows an alert; a low-risk station does not.
FR-16	The system shall display the current tank level per station and fuel type.	UN-01; RS FR3	Must	Displayed levels match the data set.
Demand forecasting				
FR-17	The system shall allow the manager to select a station and a fuel type.	UN-02; RS FR4	Must	Selecting a station and fuel type loads the matching forecast.
FR-18	The system shall produce a daily demand forecast in litres for the next 7 days for the selected station and fuel type.	UN-02; RS FR5	Must	Exactly 7 daily values are shown, starting from the next day.
FR-19	The system shall use historical sales, previous orders, current tank level, weather, local events and seasonality as forecast inputs.	UN-02; RS FR6; AS-02	Must	Changing one input in the sample data changes the forecast or its factor breakdown.
FR-20	The system shall display the forecast as a chart.	UN-02; RS FR7	Must	The station page shows a chart with the 7 forecast values.
FR-21	The system shall display a confidence level for each forecast.	UN-03; RS FR8	Must	Every forecast value has a visible confidence level.
FR-22	The system shall display the factors that influenced each forecast and their relative contribution.	UN-03; RS FR9	Must	Each forecast shows at least its main contributing factors with their share.
FR-23	The system shall flag a forecast as uncertain when its confidence is below 70%.	UN-03; RS FR10; BR-27	Must	A forecast below the threshold shows an "uncertain" flag on the station page and in the plan.
FR-83	The system shall flag a station and fuel type for manual planning when no forecast can be produced, and shall recommend no quantity for it.	UN-03; AI-21	Must	A station with no sales history shows the manual-planning flag and receives no quantity.
FR-85	The system shall display daily forecast totals per fuel type for the whole network and for each warehouse; the dispatcher sees the per-warehouse totals only.	UN-17	Must	Totals equal the sum of the station forecasts; the dispatcher sees no station-level forecast.
Urgency and priority				

ID	Requirement	Source	Priority	Acceptance Criteria
FR-24	The system shall calculate a station risk level (high, medium or low) for each station and fuel type from its time to stockout (BR-26).	UN-04; RS FR11; BR-26	Must	Every station and fuel type has one of the three levels.
FR-25	The system shall calculate a delivery priority for each station that needs fuel, using station risk, quantity to deliver, number of stations in need, distance from the warehouse, available tankers and warehouse stock.	UN-04; RS FR12; AS-07	Must	Stations in the plan are ordered by priority; changing one input changes the ranking when it should.
FR-26	The system shall allocate limited tankers, fuel and time across stations using yield management principles.	UN-04; RS FR13	Should	With resources below total demand, allocation follows the documented yield management rule.
FR-27	The system shall display the reasons for each station's urgency level.	UN-04; RS FR14	Must	Every station in the plan shows at least one reason for its level.
Delivery planning				
FR-28	The system shall recommend the litres to deliver per station and fuel type.	UN-06; RS FR21	Must	Each station in the plan shows a recommended quantity per fuel type.
FR-29	The system shall not recommend a quantity above the station's storage capacity minus its current stock.	UN-06; RS FR21; BR-04	Must	No recommendation in a test set exceeds free storage capacity.
FR-30	The system shall recommend quantities that keep each station above its safety-stock level over the planning period, when resources allow.	UN-06; RS FR21; BR-05	Must	With enough resources, no station falls below safety stock in the simulated period.
FR-31	The system shall not assign a tanker more litres than its capacity, or a compartment more than its size, in a recommended plan.	UN-06; RS FR22; BR-06, BR-31	Must	Each tanker's total load in any recommended plan is at or below its capacity, and no compartment is over its size or holds two fuel types.
FR-32	The system shall schedule recommended deliveries within each station's delivery window.	UN-06; RS FR22; BR-07	Must	Every estimated arrival time in a recommended plan falls inside the station's window.
FR-33	The system shall not recommend delivering more of a fuel type than the stock of that fuel type at the warehouse serving the stations.	UN-06; RS FR12, FR15; BR-08	Must	For each warehouse, the plan total per fuel type is at or below its uploaded stock.
FR-34	The system shall serve each station from its closest warehouse.	UN-06; stakeholder input; BR-17	Must	In a test plan, every station is linked to the warehouse with the shortest distance.
FR-35	The system shall assign the stations that need fuel to the available tankers, each with an available driver, and produce one optimal multi-stop route (one trip) per tanker for the delivery day.	UN-06; RS FR23, FR24; stakeholder input; BR-23, BR-24	Must	With N tankers and drivers available, the plan contains up to N routes and every served station is assigned to exactly one route.
FR-36	The system shall start and end each tanker's route at the tanker's home warehouse.	UN-06; stakeholder input; BR-18; AS-06; BR-25	Must	The first and last point of every route is the home warehouse of its tanker; the return leg counts in distance, time and cost.
FR-37	The system shall always produce a delivery plan when no critical resource is missing, even if resources cannot cover every station.	UN-06; stakeholder input; BR-20	Must	With fewer tankers than needed, a plan is still produced.

ID	Requirement	Source	Priority	Acceptance Criteria
FR-38	The system shall serve the highest-priority stations first when resources are short, and list the stations left unserved with their risks.	UN-06, UN-10; stakeholder input	Must	With resources short, unserved stations are the lowest-priority ones and each one shows its risk.
FR-39	The system shall not produce a plan when a critical resource is missing (no tankers, no drivers, or no stock at the serving warehouse) and shall name the missing resource.	UN-05, UN-06; stakeholder input; BR-20	Must	Removing all tankers, all drivers or a warehouse's stock stops planning with a message naming the missing resource.
Route cost				
FR-40	The system shall calculate the fuel cost of each route from its distance, the tanker's fuel consumption rate and the public fuel price, in US dollars.	UN-07; RS FR25; BR-16; OQ-04	Must	For a test route, displayed cost = distance × consumption rate × fuel price.
FR-41	The system shall include route cost in route optimization together with urgency, delivery windows and stockout risk.	UN-07; RS FR26	Must	Between two feasible plans with equal risk, the recommended one has the lower cost.
FR-42	The system shall display the cost of each route and the total cost of all routes.	UN-07; RS FR27	Must	The planning page shows each route cost and a total equal to their sum.
FR-43	The system shall display, for an edited plan, the optimal cost, the new cost and the difference in amount and percentage.	UN-07, UN-10; RS FR28	Must	After an edit, the three values appear and the difference is correct.
FR-44	The system shall show the cost change and the risks together when an edit lowers cost but creates risks.	UN-10; RS FR29	Must	A cheaper but risky edit shows both the saving and the risk list.
Route map				
FR-45	The system shall display each route on the map as a directed path with arrows.	UN-08; RS FR30	Must	Arrows on every route follow the stop order.
FR-46	The system shall number each station by its stop order in its route.	UN-08; RS FR31	Must	Numbers match the stop order in the plan.
FR-47	The system shall color each station marker by urgency: red for high, orange for medium, green for low.	UN-08; RS FR32; BR-09	Must	Marker colors match the urgency levels in the data.
FR-48	The system shall show urgency with a label or icon in addition to color.	UN-08; RS NFR9	Must	Urgency can be identified on a grayscale screenshot.
FR-49	The system shall show only the station name and ID on the map by default.	UN-08; RS FR33; stakeholder input	Must	No quantities are visible until a station is selected.
FR-50	The system shall show a details box next to a station when the user selects it, with its stop order and litres per fuel type.	UN-08; RS FR34; stakeholder input	Must	Selecting a station opens the box with correct values; selecting elsewhere closes it.
FR-51	The system shall label each route segment with its distance and travel time.	UN-08; RS FR35	Must	Every segment has a distance and time label.
FR-52	The system shall display the total distance, duration, litres and cost of each route and of the whole plan.	UN-08; RS FR36	Must	Totals equal the sum of the segments and stops.
FR-53	The system shall draw each tanker's route in its own line color, never red, orange or green.	UN-08; RS FR37; BR-10	Must	Each route has a distinct color and none uses the urgency colors.
FR-54	The system shall allow the user to show all routes or only one tanker's route.	UN-08; RS FR37	Must	Selecting a tanker hides the other routes; "all" shows them again.

ID	Requirement	Source	Priority	Acceptance Criteria
Route editing				
FR-55	The system shall allow the manager to reorder stops within a route by drag and drop.	UN-09; RS FR38; stakeholder input	Must	Dragging a stop to a new position changes the stop numbers accordingly.
FR-56	The system shall allow the manager to move a station from one tanker's route to another by drag and drop.	UN-09; RS FR39; stakeholder input; BR-21	Must	The station disappears from the first route and appears in the second; a move that overloads a tanker is accepted and raises a "capacity exceeded" risk.
FR-57	The system shall recalculate the duration, distance, arrival time per station and cost of every affected route after a reorder or a move.	UN-09, UN-10; RS FR39, FR41; BR-12	Must	Updated values appear within the time set in NFR-01 and are correct for a test case.
FR-58	The system shall allow the manager to edit the delivery quantity per station and fuel type in a plan.	UN-09; RS FR40; BR-22, BR-28	Should	Edited values are saved in the draft plan.
FR-59	The system shall generate a new optimal plan when delivery quantities are edited.	UN-09; RS FR42; BR-12	Should	After such an edit, a new plan is produced that respects BR-04 to BR-08.
FR-60	The system shall provide a "Reset to optimal" action that restores the recommended plan.	UN-09; RS FR43	Must	After reset, the plan matches the original recommendation exactly.
FR-75	The system shall display, while the manager edits a quantity, the maximum allowed for that stop and the limit that sets it (free compartment space, free tank space or warehouse stock).	UN-09, UN-10; BR-28	Should	For each of the three limits, the hint shows the correct maximum and names the limit.
FR-76	The system shall disable saving and show an error under the quantity field when the entered quantity is above the maximum, naming the limit and the maximum.	UN-09; BR-28	Should	Entering the maximum + 1 L disables saving and shows the right message; a forced request is refused.
Risk analysis				
FR-61	The system shall compare the edited plan with the optimal plan after each edit.	UN-10; RS FR44	Must	A comparison panel updates after every edit.
FR-62	The system shall detect these risks: late arrival compared with the optimal plan, stockout before arrival, missed delivery window, tanker or compartment capacity exceeded, and extra distance, time or cost.	UN-10; RS FR45	Must	Each risk type is raised in a test scenario built to trigger it, and not raised otherwise.
FR-63	The system shall give each risk a severity level: low, medium or high.	UN-10; RS FR46	Must	Every risk shown has exactly one severity level: capacity exceeded, stockout before arrival and missed window are high; late arrival is low up to 60 min, medium up to 180 min and high beyond; extra distance, time or cost is low under 10%, medium from 10% to 25% and high above 25%.
FR-64	The system shall give each risk a reason, including why the edited version was not chosen as optimal.	UN-10; RS FR46, FR47	Must	Every risk has a reason that names the affected station, route or value.

ID	Requirement	Source	Priority	Acceptance Criteria
FR-65	The system shall highlight the stations and segments affected by each risk on the map.	UN-10; RS FR48	Must	Selecting a risk highlights the matching elements on the map.
FR-66	The system shall state clearly when an edit creates no risk.	UN-10; RS FR50	Must	A no-risk edit shows an explicit "no risk" message.
FR-67	The system shall allow approval of a plan that has risk warnings, except a “capacity exceeded” risk.	UN-09; RS FR49; BR-11, BR-21	Must	A plan with medium-risk warnings can be approved; a plan with an overloaded tanker cannot.
FR-77	The system shall block approval of the plan, and of any action on the overloaded tanker, while a tanker or compartment is over capacity, and shall name the tanker and the excess litres.	UN-11; BR-21	Must	With a tanker overloaded by 1 L, approval is disabled and refused; removing the overload enables it again.
Recommendations and approval				
FR-68	The system shall list all recommended actions for the delivery day.	UN-11; RS FR51	Must	The list contains every station delivery and route of the plan.
FR-69	The system shall allow the manager to approve or reject each recommended action.	UN-11; RS FR52; BR-30	Must	Each action can be approved or rejected and shows its status.
FR-70	The system shall allow the manager to approve or reject an edited plan.	UN-11; RS FR52	Must	An edited plan can be approved and its final version is stored.
FR-71	The system shall not mark any action as final unless a manager has approved it.	UN-11; RS FR53; BR-01	Must	No action reaches "approved" status without a manager approval record.
FR-80	The system shall allow the manager to approve or reject all pending actions of a plan in one step.	UN-11; BR-30	Must	“Approve all” on a 5-action plan approves all 5 and logs each decision.
FR-81	The system shall make the “Adjust” control of an action open the delivery planner with that station selected.	UN-09	Must	Clicking Adjust opens the planner with that station selected and its route visible.
FR-82	The system shall generate a new optimal plan without a station when the manager rejects its delivery; unchanged approved actions keep their approval and changed ones return to pending.	UN-11; BR-30	Must	After a rejection, a new plan without that station is produced, freed capacity can serve other stations, and approvals follow the rule.
Decision history				
FR-72	The system shall record each approval and rejection with the user, date and time, the original recommendation, whether the approved plan is the suggested plan or an edited version, the edits performed, the final version and the risks shown.	UN-12; RS FR54; stakeholder input; AS-11	Must	Each decision creates one complete log entry, including whether the plan was edited and which edits were made.
FR-73	The system shall allow users of all three roles to view the decision history, read-only, in chronological order.	UN-12; RS FR55	Must	Past decisions are listed newest first with all recorded fields.
Data				
FR-74	The system shall run on a sample data set with simulated forecasts, routes and costs.	UN-16; RS FR56; AS-02	Must	All Must requirements can be demonstrated without real distributor data.

7. Non-Functional Requirements

ID	Category	Requirement	Source	Acceptance Criteria
NFR-01	Performance	The system shall show recalculated route values within 1 second after a reorder or move.	RS NFR4	Measured on the sample data set (20 stations, 3 warehouses, 6 tankers): 95% of edits update in ≤ 1 s.
NFR-02	Performance	The system shall load the station forecast page and the delivery plan within 3 seconds.	RS NFR5	95% of loads complete in ≤ 3 s on the sample data set.
NFR-03	Availability	The system shall be available during planned test and demonstration sessions.	Assumption (AS-13)	No unplanned downtime during scheduled sessions. A production target is open (OQ-17).
NFR-04	Scalability	The system shall support adding stations, warehouses, tankers and drivers through data, without code changes.	RS NFR10	Adding a station, warehouse, tanker or driver to the data makes it usable in planning.
NFR-05	Security	The system shall enforce role-based access on every function and data request.	RS NFR7	Direct requests to functions of another role are refused.
NFR-06	Security	The system shall not store passwords in readable form.	Assumption (AS-09)	Stored credentials cannot be read as plain text.
NFR-07	Privacy	The system shall store personal data only for user accounts (name, login, role) and driver name and ID.	Assumption (AS-09); BR-15	The data model contains no personal data other than user accounts and driver name and ID.
NFR-08	Usability	A non-technical manager shall be able to generate, edit and approve a plan without training.	RS NFR1	In a usability test, test users complete UC-05 to UC-07 unaided. At least 4 of 5 test users succeed.
NFR-09	Usability	The system shall write all explanations, reasons and risks in plain language.	RS NFR2, NFR3	A review confirms no unexplained technical terms in explanation texts.
NFR-10	Accessibility	The system shall never use color as the only way to convey information.	RS NFR9	All urgency and risk information is readable in grayscale.
NFR-11	Accessibility	Text and essential map elements shall meet WCAG 2.1 AA contrast.	Assumption (AS-10)	A contrast checker reports no AA failures on the main screens.
NFR-12	Maintainability	The forecasting, priority, routing and cost modules shall be replaceable without changes to the user interface.	RS NFR11	A simulated module can be swapped for another implementation with the same interface.
NFR-13	Reliability	The system shall give the same results for the same inputs.	RS NFR6	Running the same plan twice gives identical forecasts, priorities, routes and costs.
NFR-14	Reliability / Auditability	The system shall not allow decision log entries to be edited or deleted.	RS NFR8; BR-13	No user role, including admin, can change or remove a log entry.
NFR-15	Interoperability	The system shall receive external data (weather, events, map distances) through defined interfaces that can be connected to real sources later.	RS NFR11; Project Scope 9; AS-08	Each data source is accessed through one documented interface.
NFR-16	Interoperability	The system shall run on current versions of major desktop browsers.	RS NFR12	Main use cases work on the latest Chrome, Firefox and Edge.
NFR-17	Compliance	The system shall not set pump prices or control any fuel system.	RS NFR13; BR-14	No function changes prices or sends commands to equipment.

ID	Category	Requirement	Source	Acceptance Criteria
NFR-18	Compliance	Recommended plans shall respect storage capacity and tanker capacity limits.	BR-04, BR-06; regulators' concern	No recommended plan breaks either limit. Other regulatory rules: OQ-13.
NFR-19	Performance	The system shall generate a full optimal plan within 10 seconds.	Stakeholder input	≤ 10 s on the sample data set.
NFR-20	Performance	The system shall load the network overview and the decision history within 3 seconds.	Assumption	95% of loads complete in ≤ 3 s.
NFR-21	Security	The system shall lock an account for 15 minutes after 5 failed logins within 15 minutes.	Assumption	The 6th failed attempt within 15 min is refused.
NFR-22	Security	The system shall end a session after 30 minutes without activity.	Stakeholder input	A session idle for 30 min requires a new login.
NFR-23	Security	The system shall use HTTPS for all traffic outside a local machine.	Assumption	No unencrypted request is accepted in a deployed test.
NFR-24	Scalability	The system shall handle 100 stations, 10 warehouses and 30 tankers.	Assumption	Plan generation ≤ 30 s; 95% of edits update in ≤ 2 s.
NFR-25	Privacy	The system shall keep passwords, tokens and session IDs out of its logs.	Assumption	A log scan finds none.
NFR-26	Reliability	The system shall make every save all-or-nothing.	Assumption	A failed save leaves the previous data unchanged.
NFR-27	Accessibility	The system shall offer a keyboard alternative for every function, including drag and drop.	Assumption	UC-03 to UC-07 can be completed with the keyboard only.
NFR-28	Maintainability	The forecasting, priority, routing, cost and editing modules shall have automated tests.	Assumption	≥ 80% line coverage on these modules.
NFR-29	Observability	The system shall log requests, plan generations, edits, decisions and errors.	Assumption	100% of these events are logged with time, user and request ID.
NFR-30	Portability	The system shall start with one command on Windows, macOS and Linux.	Assumption	A fresh machine runs the platform with one command.
NFR-31	Usability	The system shall show litres as whole numbers with thousands separators, times in 24-hour format and costs in US dollars.	Stakeholder input	A UI review finds no other format.

8. AI-Specific Requirements

AI is involved in forecasting, priority scoring, route optimization and explanations. In v1 these are simulated, but the requirements below define how the AI must behave when real models are used.

8.1 AI capabilities

- Demand forecasting: predicts daily litres per station and fuel type for 7 days, with daily totals per warehouse and for the network (simulated in v1).
- Risk and priority scoring: assigns station risk and delivery priority.
- Route optimization: assigns stations to tankers and orders stops under tanker and compartment capacity, window, risk and cost limits. This is an optimization method, not necessarily machine learning.
- Risk analysis and explanation: compares edited plans with the optimal plan and explains forecasts, priorities and risks.

8.2 Inputs and outputs

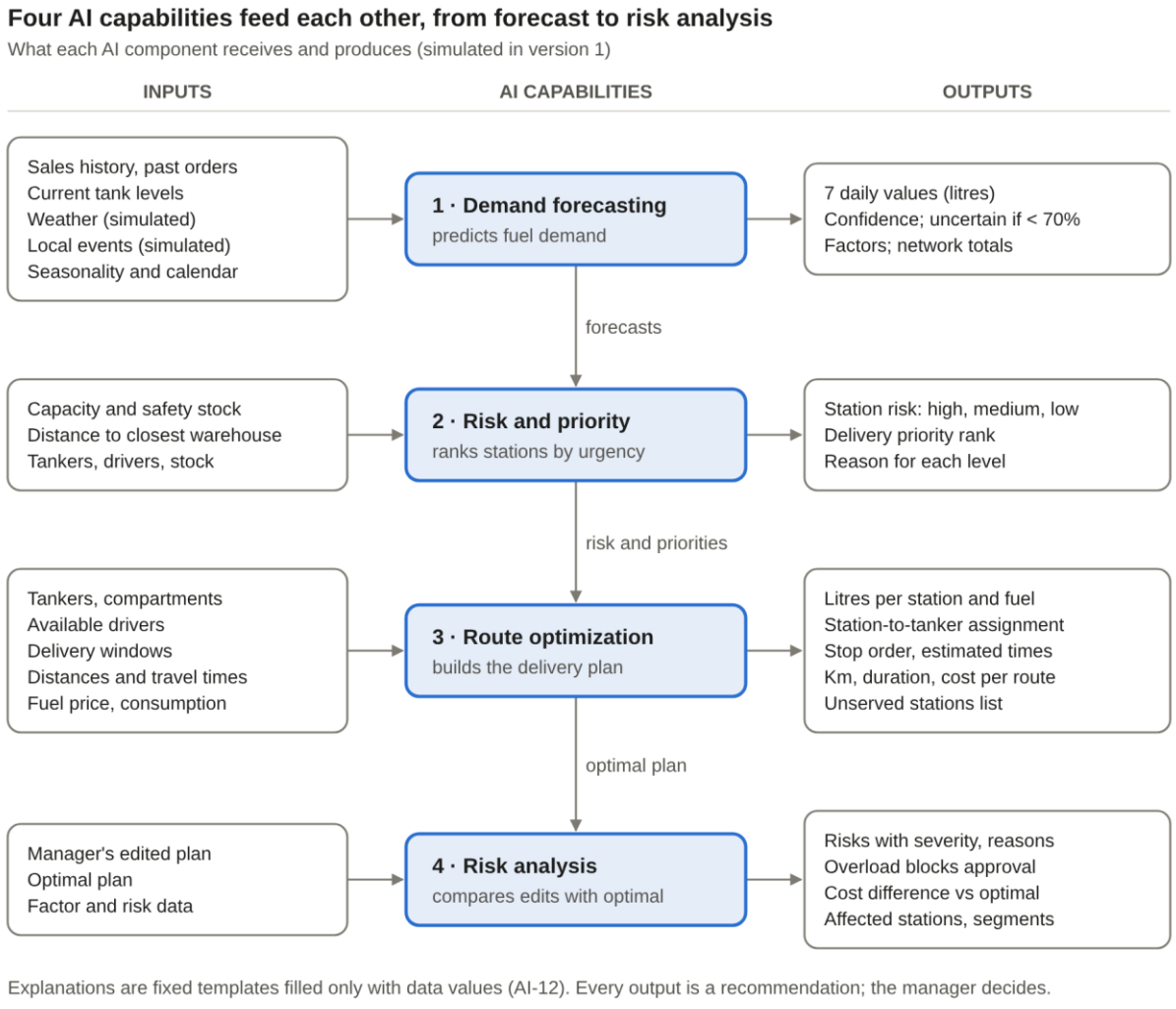


Figure 2 – AI inputs and outputs

8.3 Model responsibilities

- **AI-01:** The forecasting component shall produce each forecast together with its confidence level and its contributing factors.
- **AI-02:** The priority component shall produce a risk level and a priority for each station, with reasons.
- **AI-03:** The routing component shall produce only plans that respect BR-04 to BR-08.
- **AI-04:** The AI components shall only recommend. Decisions belong to the operations manager.

8.4 Human-in-the-loop requirements

- **AI-05:** Every recommendation shall require manager approval before it becomes final (BR-01).
- **AI-06: The manager shall be able to override a recommendation's stop order, tanker assignment and quantities; the operational resources stay as uploaded by the dispatcher.**
- **AI-07:** Every override shall be logged with the system recommendation it replaced (FR-72).

8.5 Accuracy expectations

- Accuracy target for the version 1 pipeline check: MAPE ≤ 15% on held-back sample data.
- **AI-08:** Forecast accuracy shall be measured with Mean Absolute Percentage Error (MAPE, the average % gap between forecast and actual) on held-back sample data. The version 1 target is MAPE ≤ 15%.
- In v1, forecasts are simulated, so accuracy shows that the pipeline works, not real-world performance.

8.6 Confidence and uncertainty handling

- **AI-09:** Every forecast shall show a confidence level (FR-21).
- **AI-10:** Forecasts below the confidence threshold shall be flagged as uncertain (FR-23, BR-27).
- **AI-11:** A recommendation based on an uncertain forecast shall carry the same flag in the plan and on the recommendations page.

8.7 Hallucination mitigation

- **AI-12:** Every explanation shall be built only from the actual input data and calculated values. It shall not state facts, numbers or events that are not in the data.
- **AI-13: Explanations shall be built from fixed sentence templates filled only with data values; no text-generating model (LLM) is used in version 1.**

8.8 Explainability

- **AI-14:** Forecasts shall show their main factors and relative contribution (FR-22).
- **AI-15:** Urgency levels, risks and the "why not optimal" comparison shall show their reasons (FR-27, FR-64).

8.9 Bias and fairness considerations

- **AI-16:** A station shall not get a lower priority only because it is remote or has low sales volume when its stockout risk is high. If resources force this, the reason shall be shown.
- **AI-17:** The sample data shall cover different seasons, station sizes and locations, so the forecasts are not tuned to one type of station.

8.10 Data requirements

- Historical daily sales per station and fuel type (at least 365 days).
- Previous orders: past deliveries per station and fuel type.
- Station data: location, tank capacity, safety-stock level, current level and delivery window (maintained by the system admin or the operations manager).
- Weather, local events and calendar or seasonal data.
- Operational resources: tankers (compartment sizes, capacity, fuel consumption rate, home warehouse), drivers and other activity data (OQ-20).
- Fuel price: public price, in US dollars (consumption rates: OQ-04).
- Warehouse locations and stock per warehouse and fuel type.
- Distances and travel times between all points (AS-08).
- Unloading times per delivered quantity (sample values in version 1; real data later from data managers).

8.11 Model monitoring

- **AI-18:** Each plan shall record the input data it was built from (data version and the operational resources uploaded at that time) and the version of each model used, so any recommendation can be traced and explained later.
- **AI-19 (future, with real data):** The system shall compare forecasts with actual sales and track the error over time.

8.12 Model failure handling

- **AI-20:** If an input is missing, the forecast shall be flagged low-confidence and the missing input named.
- **AI-21:** If no forecast can be produced for a station, the station shall be flagged for manual planning instead of receiving a recommendation.
- **AI-22:** The system shall always produce a plan, even when resources cannot cover every station: it serves the highest-priority stations first and lists the unserved stations with their risks. It shall only stop when a critical resource is missing (no tankers, no drivers or no warehouse stock), and shall name it.

8.13 AI safety constraints

- **AI-23:** The system shall never execute actions on its own (BR-01).
- **AI-24:** The system shall never set prices or control fuel systems (BR-14).
- **AI-25:** Recommended plans shall never exceed storage, tanker or compartment capacity, even when resources are short (BR-04, BR-06).

8.14 Privacy requirements

- **AI-26:** The AI components shall use operational data only, with no personal data.

8.15 Evaluation criteria

- Forecast error (MAPE) on held-back sample data. Target: ≤ 15% in version 1.
- 0 constraint violations (BR-04 to BR-08) in recommended plans.
- Live recalculation within NFR-01.
- 100% of forecasts, priorities and risks have an explanation.
- Usability test results for UC-05 to UC-07 (NFR-08).

9. User Stories

ID	User Story	Need
US-01	As an operations manager, I want to see all stations on a map with stockout alerts, so that I know at a glance where action is needed.	UN-01
US-02	As an operations manager, I want to see a 7-day demand forecast for a station and fuel type, so that I can plan deliveries ahead.	UN-02
US-03	As an operations manager, I want to see how confident each forecast is and what influenced it, so that I know how much to trust it.	UN-03
US-04	As an operations manager, I want to see which stations are most urgent and why, so that I serve the right stations first.	UN-04
US-05	As a dispatcher, I want to upload the operational resources available for the day (tankers, drivers, warehouse stock), so that the plan is based on real resources.	UN-05
US-06	As an operations manager, I want to get recommended quantities and one route per available tanker, so that I don't have to build the plan by hand.	UN-06
US-07	As an operations manager, I want to see the fuel cost of each route, so that I can keep delivery costs low.	UN-07
US-08	As an operations manager, I want to see only station names and IDs on the map and open details on demand, so that the map stays readable.	UN-08
US-09	As an operations manager, I want to drag and drop stops to reorder them or move them to another tanker, so that I can adapt the plan to what I know on the ground.	UN-09
US-10	As an operations manager, I want to see time, distance, arrival times and cost update live as I edit, so that I see the impact of my change immediately.	UN-09, UN-10
US-11	As an operations manager, I want to see the risks of my edit, how serious they are and why, so that I can decide knowingly.	UN-10
US-12	As an operations manager, I want to reset my edits to the optimal plan, so that I can start over quickly.	UN-09
US-13	As an operations manager, I want to approve or reject each recommendation, so that nothing happens without my decision.	UN-11
US-14	As a distributor manager, I want to see a history of all decisions, so that I can trace who decided what and when.	UN-12
US-15	As a system admin, I want to manage user accounts and roles, so that only the right people can do each action.	UN-13

ID	User Story	Need
US-16	As a system admin, I want to manage station records, so that the network data stays current.	UN-14
US-17	As a dispatcher, I want to see the total forecast demand per warehouse and fuel type, so that the warehouse can prepare stock without adding up station forecasts.	UN-17
US-18	As an operations manager, I want to see the maximum quantity allowed while I edit a delivery, so that I never overload a tanker or a tank.	UN-09, UN-10
US-19	As a dispatcher, I want to record when each tanker leaves and returns, so that planned and actual trip times can be compared.	UN-05, UN-18
US-20	As an operations manager, I want to approve or reject the whole plan in one step, so that I save time when the plan is fine.	UN-11
US-21	As an operations manager, I want to manage station records, so that the network data stays current.	UN-14

10. Use Cases

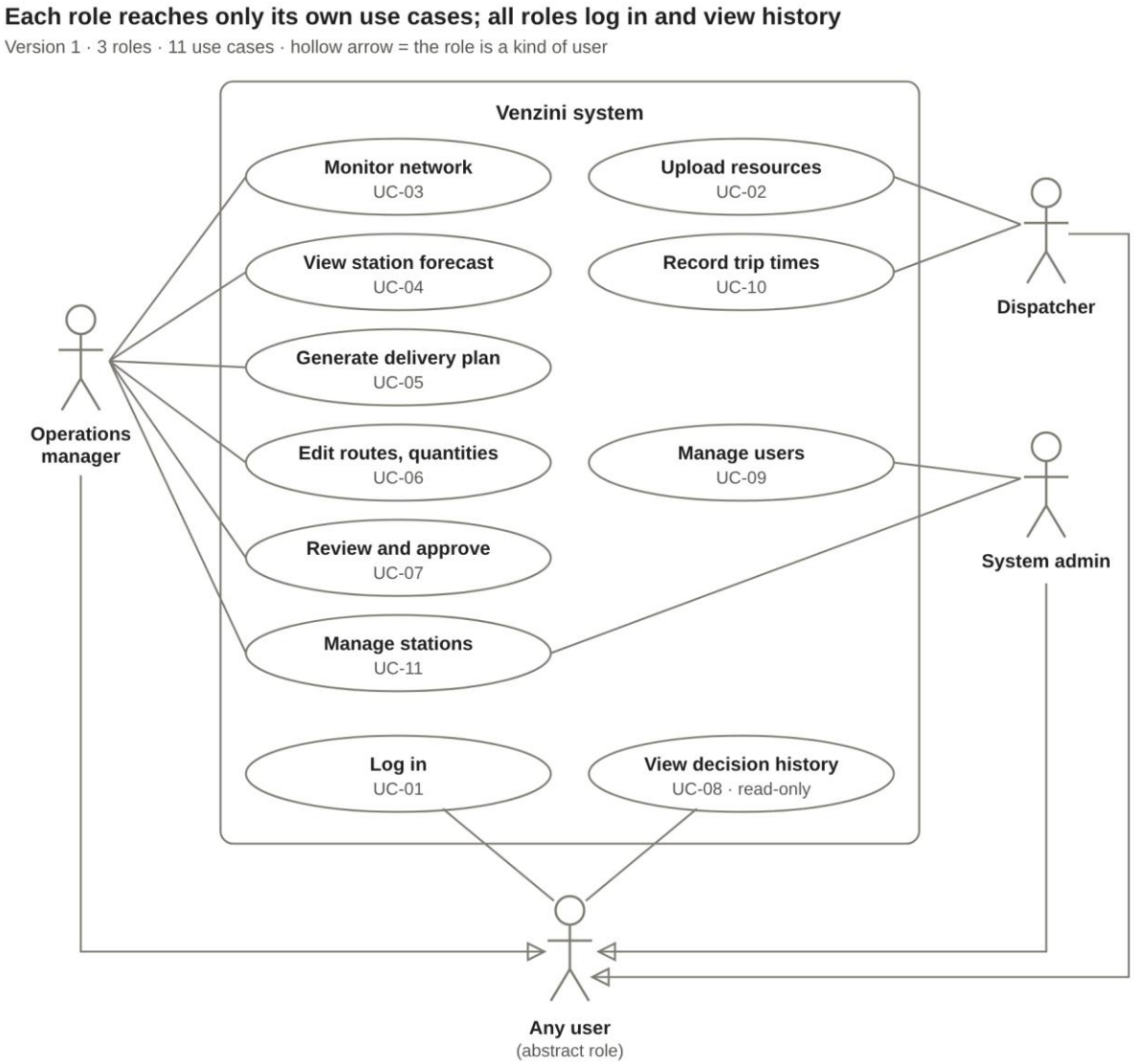


Figure 3 – Use case diagram

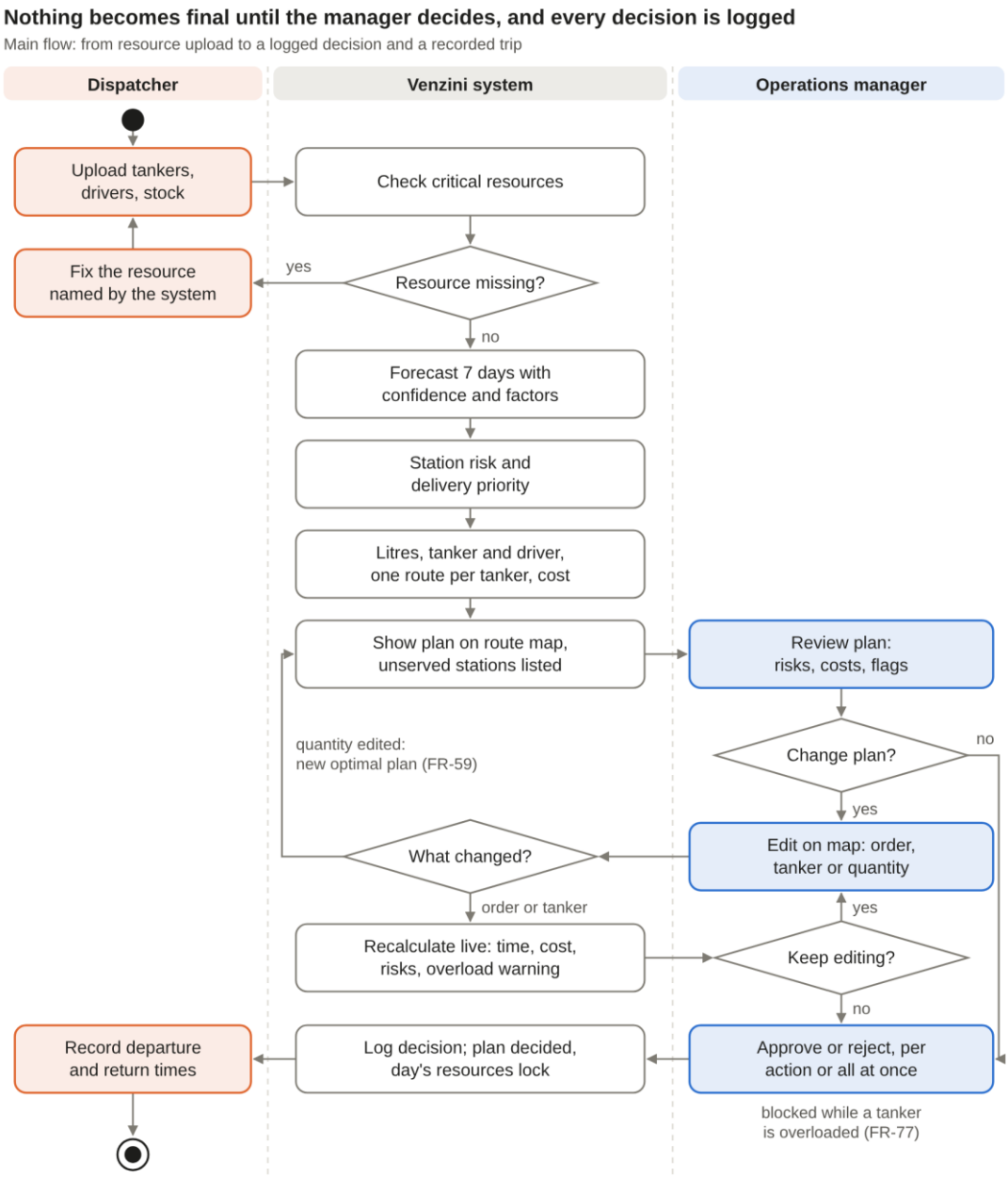


Figure 4 – Main planning flow (activity diagram)

UC-01 – Log in

Actor	Any user
Preconditions	User has an active account.
Trigger	User opens the platform.
Main flow	1. User enters login and password. 2. System checks the credentials. 3. System opens the home page for the user's role.
Alternative flows	– None.
Exceptions	– Wrong credentials: system shows an error and stays on the login page. – Deactivated account: system refuses access.
Postconditions	User is logged in with the permissions of their role.

UC-02 – Upload resources

Actor	Dispatcher
Preconditions	Dispatcher is logged in. The plan for the day is not yet decided.
Trigger	Resources for a delivery day must be recorded or changed.
Main flow	1. Dispatcher opens the resources page and selects the delivery day. 2. Dispatcher enters available tankers, their compartment sizes, capacities and home warehouses. 3. Dispatcher enters available drivers. 4. Dispatcher enters the fuel stock of each warehouse per fuel type. 5. Dispatcher saves. 6. System validates and stores the data, with user and timestamp.
Alternative flows	– Resources already exist for the day: dispatcher edits and saves them.
Exceptions	– Invalid value (e.g., negative capacity): system rejects it and names the field. – Plan already decided: system refuses the change (AS-05).
Postconditions	Current operational resources for the day are stored and visible to the manager.

UC-03 – Monitor network

Actor	Operations manager
Preconditions	Manager is logged in.
Trigger	Manager opens the network overview.
Main flow	1. System shows all active stations on the map. 2. System shows tank levels and stockout alerts. 3. Manager selects a station to open its details.
Alternative flows	– No station at high risk: system shows no alert.
Exceptions	– Station data missing: system marks the station as "data unavailable".
Postconditions	Manager knows which stations need attention.

UC-04 – View station forecast

Actor	Operations manager
Preconditions	Manager is logged in. Historical data exists for the station.
Trigger	Manager selects a station and fuel type.
Main flow	1. System produces the 7-day forecast. 2. System shows the chart, confidence level and contributing factors.
Alternative flows	– Low confidence: system flags the forecast as uncertain.
Exceptions	– Forecast cannot be produced: system explains why and flags the station for manual planning (AI-21).
Postconditions	Manager sees the expected demand and how reliable it is.

UC-05 – Generate delivery plan

Actor	Operations manager
Preconditions	Resources for the day are uploaded. Forecasts are available.
Trigger	Manager opens the delivery planner for a day.

Main flow	1. System calculates station risk and delivery priority. 2. System links each station to its closest warehouse. 3. System recommends litres per station and fuel type. 4. System assigns stations to tankers (each with a driver) and builds one route (one trip) per tanker, from and back to its home warehouse, filling compartments by fuel type. 5. System calculates the cost and estimated trip times of each route. 6. System shows the plan on the route map.
Alternative flows	– Some forecasts are uncertain: matching recommendations carry the uncertain flag. – Resources are short: system serves the highest-priority stations first and lists the unserved stations with their risks.
Exceptions	– No resources uploaded: system asks for resources first. – Critical resource missing (no tankers, no drivers or no warehouse stock): system names it and does not produce a plan (AI-22).
Postconditions	An optimal draft plan is ready for review.

UC-06 – Edit routes and quantities

Actor	Operations manager
Preconditions	A draft plan exists (UC-05).
Trigger	Manager drags a stop or station on the map, or edits a delivery quantity.
Main flow	1. Manager reorders a stop within a route, or moves a station to another tanker. 2. System recalculates time, distance, arrival times and cost of affected routes. 3. System compares with the optimal plan and shows risks, severity, reasons and cost difference. 4. System highlights affected stations and segments.
Alternative flows	– No risk created: system says so clearly. – Manager clicks "Reset to optimal": system restores the recommended plan. – Manager edits a quantity (rest of version 1): system shows the allowed maximum, blocks values above it, then builds a new optimal plan.
Exceptions	– Receiving tanker over capacity: system raises a "capacity exceeded" risk and blocks approval until the overload is removed (BR-21).
Postconditions	An edited draft plan exists with its risks and costs visible.

UC-07 – Review and approve

Actor	Operations manager
Preconditions	A draft plan exists (original or edited).
Trigger	Manager opens the recommendations page.
Main flow	1. System lists all recommended actions with reasons, risks and costs. 2. Manager approves or rejects each action, or the whole plan. 3. System records each decision (FR-72).
Alternative flows	– Manager clicks Adjust on an action: system opens the planner on that station. – Manager rejects a station: system builds a new optimal plan without it; unchanged approvals stay, changed actions return to pending.
Exceptions	– Plan has risk warnings: approval is still allowed (BR-11). – A tanker is overloaded: approval is blocked until it is fixed (BR-21).
Postconditions	Each action is approved or rejected and logged; the day’s resources lock once the plan is decided.

UC-08 – View decision history

Actor	Operations manager, dispatcher, system admin
Preconditions	User is logged in.
Trigger	User opens the decision history.
Main flow	1. System lists past decisions newest first. 2. User opens one entry to see the original recommendation, the final version, changes and risks.
Alternative flows	– No decisions yet: system shows an empty list message.
Exceptions	– None.
Postconditions	User has traced past decisions. Nothing is changed.

UC-09 – Manage users

Actor	System admin
Preconditions	Admin is logged in.
Trigger	Admin opens the administration page.
Main flow	1. Admin creates, edits or deactivates a user and assigns a role. 2. System saves the change.
Alternative flows	– None.
Exceptions	– Duplicate login: system rejects it.
Postconditions	User accounts are up to date.

UC-10 – Record trip times

Actor	Dispatcher
Preconditions	Dispatcher is logged in. The day's plan is decided, with at least one approved trip.
Trigger	A tanker leaves or returns to its home warehouse.
Main flow	1. Dispatcher opens the trips page and selects the trip. 2. Dispatcher enters the actual departure time. 3. When the tanker returns, dispatcher enters the actual return time. 4. System validates and stores the times and shows the gap with the estimated duration.
Alternative flows	– Wrong time entered: dispatcher corrects it before the return is recorded.
Exceptions	– Return before departure: system rejects it and names the field.
Postconditions	The trip is marked "departed" or "returned", and its estimated vs. actual gap is stored.

UC-11 – Manage stations

Actor	System admin, operations manager
Preconditions	User is logged in as system admin or operations manager.
Trigger	User opens the stations page.
Main flow	1. User creates, edits or deactivates a station, with its location, tanks per fuel type and delivery window. 2. System validates and saves the change.
Alternative flows	– None.

Exceptions	– Duplicate station ID or invalid value (e.g., safety stock above capacity): system rejects it and names the field.
Postconditions	The station list is up to date; a deactivated station no longer appears in new plans.

11. Business Rules

ID	Rule	Source
BR-01	No recommendation becomes final without a manager's approval.	RS FR53
BR-02	Only operations managers can approve or reject.	RS NFR7
BR-03	Only dispatchers can upload or update operational resources.	RS NFR7; stakeholder input
BR-04	A recommended quantity cannot exceed the station's storage capacity minus its current stock.	RS FR21
BR-05	Recommended plans keep stations above their safety-stock level when resources allow.	RS FR21
BR-06	A recommended tanker load cannot exceed the tanker's capacity, even when resources are short.	RS FR22; stakeholder input
BR-07	Recommended deliveries must arrive within the station's delivery window.	RS FR22
BR-08	Recommended totals per fuel type cannot exceed the stock of the serving warehouse.	RS FR12, FR15
BR-09	Urgency colors: red = high, orange = medium, green = low.	RS FR32
BR-10	Tanker route lines never use red, orange or green.	RS FR37
BR-11	Risks are warnings and do not block approval, except "capacity exceeded" (BR-21).	RS FR49
BR-12	Reordering or moving stations triggers recalculation; editing quantities triggers a new optimal plan.	RS FR41, FR42
BR-13	Decision log entries cannot be edited or deleted.	RS NFR8
BR-14	The platform never sets pump prices or controls fuel systems.	RS NFR13
BR-15	Tanker drivers have no access to the platform in any version.	Stakeholder input
BR-16	Route cost covers only the tanker's own fuel consumption.	Stakeholder input
BR-17	Each station is served from its closest warehouse. There can be several warehouses, depending on the area.	Stakeholder input
BR-18	Each tanker is tied to one home warehouse.	Stakeholder input
BR-19	A tanker may load at another warehouse only if it belongs to the same petroleum company. The process and the approver are not yet defined (OQ-07).	Stakeholder input
BR-20	The system always produces a plan unless a critical resource is missing (no tankers, no drivers or no warehouse stock).	Stakeholder input
BR-21	A plan in which any tanker or compartment load exceeds its capacity cannot be approved until the overload is removed.	Stakeholder input
BR-22	The operations manager cannot change operational resources; they may change only stop order, a station's tanker assignment and delivery quantities.	Stakeholder input
BR-23	A trip lasts from the tanker's departure from its home warehouse until its return and may span more than one day; a tanker starts at most one trip per delivery day.	Stakeholder input
BR-24	Each station receives its delivery from exactly one tanker per plan (no split delivery).	Stakeholder input
BR-25	Every route ends at the tanker's home warehouse; the return leg counts in distance, duration and cost.	Stakeholder input
BR-26	Station risk per fuel type: high if time to stockout is under 24 h, medium from 24 h to 72 h, low above 72 h.	Stakeholder input
BR-27	A forecast is uncertain when its confidence is below 70%.	Stakeholder input

ID	Rule	Source
BR-28	The maximum quantity the manager can enter for a stop is the smallest of: the free space in the tanker's compartments for that fuel type, the station's free storage, and the remaining stock of that fuel at the serving warehouse.	Stakeholder input
BR-29	The system calculates estimated trip times; only the dispatcher records actual departure and return times.	Stakeholder input
BR-30	The manager can decide each action or the whole plan. A plan is decided when every action is approved or rejected, and the day's resources lock at that moment. Rejecting a station triggers a new optimal plan.	Stakeholder input
BR-31	A tanker has compartments and can carry several fuel types on one trip; each compartment holds one fuel type and never more than its size.	Stakeholder input

12. Assumptions

This section separates what is assumed from what is fixed or still open.

12.1 Assumptions

ID	Assumption
AS-02	In v1, forecasts, routes, costs and external data (weather, events, distances) are simulated.
AS-03	The sample data is realistic enough to test and demonstrate the platform.
AS-04	The dispatcher uploads operational resources before the manager starts planning.
AS-05	Operational resources for a delivery day can no longer be changed once its plan is decided.
AS-06	In version 1, each tanker loads only at its home warehouse. Loading at another warehouse of the same company is not modelled until OQ-07 is answered.
AS-07	Version 1 core uses a simpler priority rule; full yield management comes later in version 1.
AS-08	Distances and travel times come from a map service or a simulated distance table.
AS-09	Personal data is limited to user accounts, and passwords are stored securely.
AS-10	WCAG 2.1 AA is the contrast standard.
AS-11	The decision log stores the risks shown at the time of the decision, whether the approved plan was the suggested plan or an edited one, and the edits performed.
AS-13	Availability only needs to cover test and demonstration sessions in v1.
AS-15	For now, “operational resources” covers tankers, drivers and warehouse stock. Other activity data will be added once identified (OQ-20).

12.2 Constraints

ID	Constraint
CO-01	One developer (Mohamed Zouari) builds the whole platform.
CO-02	No access to real distributor data in v1.
CO-03	v1 targets desktop browsers only; no mobile application.
CO-04	No pump pricing and no control of fuel systems.
CO-05	Every recommendation requires human approval.
CO-06	The first MVP is a clickable frontend prototype with fake data.

12.3 Open questions

ID	Question
OQ-04	Who provides each tanker’s fuel consumption rate, and how often does it change?
OQ-07	How does loading at another warehouse of the same company work, and who approves it?
OQ-11	What target values define success for BG-01 to BG-03?
OQ-13	Must the platform enforce other regulatory rules (e.g., driver working hours, hazardous-goods routes)?
OQ-15	How is an approved plan handed over to drivers and the depot (print, export, none)?
OQ-16	How long must the decision history be kept?
OQ-17	Is a production availability target needed after v1?
OQ-20	What other activity data belongs to “operational resources” besides tankers, drivers and warehouse stock?

12.4 Risks

ID	Level	Risk	Mitigation
RK-01	High	Multi-tanker route optimization with capacities and time windows is computationally hard; the 1 s live target may not hold.	Use a fast heuristic for live recalculation; test on a fixed sample size (20 stations, 6 tankers).
RK-03	Medium	Simulated data may not convince reviewers that forecasts are meaningful.	Build realistic sample data (AS-03, AI-17) and document how it was built.
RK-04	Medium	The "very flexible" goal can cause scope creep.	Keep the MVP, version 1 core and rest-of-version-1 split from the Project Scope.
RK-05	Medium	One developer carries all design, build and documentation work.	Prioritize Must requirements; move Should items after the MVP.
RK-06	Medium	Explanations may be vague or inconsistent.	Build explanations from structured factor data (AI-12).
RK-07	Medium	Trips can last several days, which makes scheduling and tanker availability harder.	One trip start per tanker per day (BR-23); a tanker is marked available again only after its return is recorded.
RK-08	Low	Re-planning after a rejection can change actions the manager already approved.	Unchanged approvals are kept; changed actions return to pending for review (BR-30).